

Green AI

Treating computational and environmental cost as a reported outcome alongside accuracy

GLOSSARY 05 - SHEET 6 / 8

PROFESSIONAL DEFINITION

Green AI is the practice of treating the computational and environmental cost of machine-learning research and deployment as a first-class evaluation criterion alongside accuracy, and of reporting it. It is contrasted with Red AI, in which accuracy improvements are pursued through increased computation without that cost being reported. The proposed reporting measure is total floating-point operations required to produce a result, which is hardware-independent and comparable across studies (Schwartz et al., 2020).

WHO DEFINED IT ORIGINALLY

Roy Schwartz, Jesse Dodge, Noah A. Smith and Oren Etzioni of the Allen Institute for AI proposed the distinction in a 2019 preprint published in Communications of the ACM in 2020. Emma Strubell, Ananya Ganesh and Andrew McCallum (2019) provided the influential energy estimates for training; Patterson et al. (2021) supplied a more precise accounting of carbon cost.

WHY THIS DEFINITION WAS DEVELOPED

Reported progress was being achieved substantially by increasing computation, with the cost invisible in publication and inaccessible to smaller institutions. Making computational cost a reported outcome was intended to address both the environmental consequence and the concentration of research capability in organisations able to afford the compute.

HOW IT IS USED

Report computational cost alongside performance for every model, and evaluate improvements against cost as well as against accuracy; where the accuracy gain per unit of computation is negligible, the smaller model is the correct deployment choice.

WHEN IT IS USED

At model selection, at procurement of AI services, in sustainability reporting, and when a marginal accuracy improvement requires a disproportionate increase in computation.

BY WHOM IT IS USED

Machine-learning practitioners and research leads, sustainability and reporting functions, procurement teams comparing vendor offerings, and boards with climate commitments.

FIGURE 1 - ACCURACY AGAINST COMPUTATIONAL COST

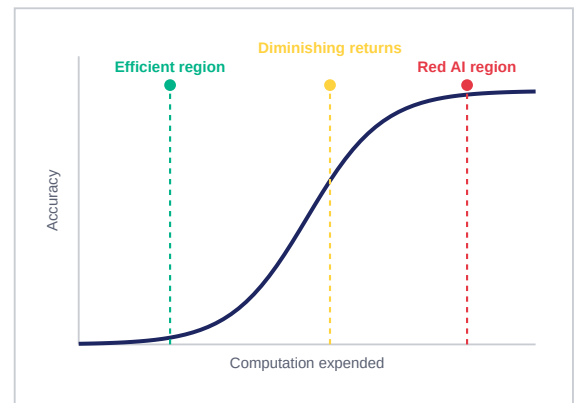


Figure 1. Accuracy rises with computation at a sharply decreasing rate. Reporting only the vertical axis makes movement into the right-hand region look like progress; reporting both axes makes it a cost decision (after Schwartz et al., 2020).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor evaluation norm reported accuracy alone, which made increased computation an invisible and therefore costless input. Green AI adds efficiency to the reported result and proposes a hardware-independent unit, which permits comparison across studies and across time in a way that energy or monetary cost does not. Its limitation is that reporting remains voluntary in most venues, so the distinction currently operates as a norm rather than a requirement.

RELATED DEFINITIONS IN THIS COURSE

Life Cycle Assessment - Unit 5

Externality - Unit 5

MLOps - Unit 4

AI Governance - Unit 6

Productivity Paradox - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.-M., Rothchild, D., So, D., Texier, M. and Dean, J. (2021) Carbon Emissions and Large Neural Network Training. arXiv:2104.10350.
- Schwartz, R., Dodge, J., Smith, N.A. and Etzioni, O. (2020) 'Green AI', Communications of the ACM, 63(12), pp. 54-63.
- Strubell, E., Ganesh, A. and McCallum, A. (2019) 'Energy and policy considerations for deep learning in NLP', in Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. Stroudsburg, PA: ACL, pp. 3645-3650.

Verification note: published energy and carbon estimates for model training vary by more than an order of magnitude depending on hardware, data-centre efficiency, grid carbon intensity and whether inference is included. Cite the primary study and its stated boundary conditions rather than a secondary summary figure.